Anomaly Detection 🕵️

This is an unsupervised learning algorithm that examines an unlabeled dataset of "normal" events and learns to detect or "flag" unusual or anomalous events.

The Problem

Imagine you are a manufacturer of aircraft engines.

- You produce many engines, and you want to ensure they are reliable.
- You can measure features for each engine, such as:
  - x_1: Heat generated
  - x_2: Vibration intensity
- **Dataset:** You have a dataset of m engines that you have manufactured. Most of these are normal, good engines.
- **Goal:** When a new engine (x_{test}) is made, you want to know: "Is this engine similar to the ones I've seen before (normal), or is it strange (anomalous)?"

How it Works: Density Estimation

The core technique used for anomaly detection is called **Density Estimation**.

1. **Build a Model ($p(x)$):**
   - The algorithm uses the training data to build a probability model $p(x)$.
   - This model tells you the probability of seeing a specific set of features.
   - Values of x that are "normal" (seen often) will have a **high probability**.
   - Values of x that are "unusual" (rare) will have a **low probability**.
2. **Set a Threshold (ϵ):**
   - You choose a small number ϵ (epsilon) as your cutoff threshold.
3. **Detect Anomalies:**
   - For a new example x_{test}, calculate $p(x_{test})$.
   - **If $p(x_{test}) < \epsilon$:** The probability is very low. Flag it as an **anomaly**.
   - **If $p(x_{test}) \geq \epsilon$:** The probability is high enough. It is **normal**.

Real-World Applications

Anomaly detection is extremely common in industry, used for ensuring quality and security.

- **Fraud Detection:**

    - **Context:** Websites tracking user behavior (login frequency, typing speed, transaction volume).
    - **Usage:** Model $p(x)$ for a user's typical behavior. If a user's activity suddenly has a very low probability (e.g., logging in from a new country and typing much faster), flag it for security review (e.g., ask for 2-factor authentication).
    - **Financial Fraud:** Identifying unusual credit card transactions.
- **Manufacturing (Quality Control):**
    - **Context:** Producing items like aircraft engines, smartphones, or circuit boards.
    - **Usage:** Measure features of every unit. If a unit has anomalous measurements (e.g., high vibration), inspect it for defects before shipping.
- **Monitoring Data Centers:**
    - **Context:** Managing thousands of computer servers.
    - **Usage:** Monitor features like memory usage, CPU load, and network traffic. If a machine behaves anomalously (e.g., CPU load spikes while network traffic is zero), it might be broken, overheating, or hacked.